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RTCA/DO-160G

Environmental Conditions and Test Procedures for Airborne Equipment

Section 4

Temperature and Altitude

Important Notice

Information contained in these sections is pertinent to all test procedures described in the other sections of this document. Further, Appendix A is applicable for identifying environmental tests performed.

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4.0 Temperature and Altitude

4.1 Purpose of the Tests

These tests determine the performance characteristics of equipment at the applicable categories for the temperatures and altitudes specified in Table 4-1 and at the pressures defined in Table 4-2.

4.2 General

Several temperature and altitude test procedures are defined, each to be selected according to the category for which the equipment is designed to be used when installed in an aircraft (see Subsection 4.3 and Table 4-1).

Note: The selection of a temperature/altitude category depends on the location in (or on) the aircraft, the maximum operating altitude of the aircraft and whether the equipment is located within a temperature and/or pressure controlled area. The above conditions must be taken into consideration by the equipment designer in evaluating these requirements, which are determined by the end application and use of the equipment.

4.3 Equipment Categories

The following categories cover the wide range of environments known to exist in the majority of aircraft types and installation locations. It should be recognized that not all possible combinations of temperatures and altitude limits are covered in these equipment categories. Categories for in-flight loss of cooling are defined in paragraph 4.5.5.

Category A1

Equipment intended for installation in a controlled temperature and pressurized location, on an aircraft within which pressures are normally no lower than the altitude equivalent of 15,000 ft (4,600 m) Mean Sea Level (MSL), is identified as Category A1. This category may also be applicable to equipment installed in temperature controlled but unpressurized locations on an aircraft that operates at altitudes no higher than 15,000 ft (4,600 m) MSL.

Category A2

Equipment intended for installation in a partially controlled temperature but pressurized location on an aircraft within which the pressures are normally no lower than the altitude equivalent of 15,000 ft (4,600 m) MSL is identified as Category A2. This category may also be applicable to equipment installed in partially controlled temperature but unpressurized locations on an aircraft that operates at altitudes no higher than 15,000 ft (4,600 m) MSL.

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Category A3

Equipment intended for installation in a controlled or partially controlled temperature but pressurized location within an aircraft within which the pressures are normally no lower than the altitude equivalent of 15,000 ft (4,600 m) MSL, where the temperatures will be more severe than those for categories A1 and A2, is identified as Category A3.

Category A4

Equipment intended for installation in a controlled temperature and pressurized location, on an aircraft within which pressures are normally no lower than the altitude equivalent of 15,000 ft. (4,600m) Mean Sea Level (MSL), for which temperature requirements differ from category A1 as declared by the equipment manufacturer. This category may also be applicable to equipment installed in a temperature controlled but unpressurized locations on an aircraft that operates at altitudes no higher than 15,000ft. (4,600m) MSL, for which temperature requirements differ from category A1 as declared by the equipment manufacturer.

Category B1

Equipment intended for installation in a non-pressurized but controlled temperature location in an aircraft that is operated at altitudes up to 25,000 ft (7,620 m) MSL is identified as Category B1.

Category B2

Equipment intended for installation in non-pressurized and non-controlled temperature locations on an aircraft that is operated at altitudes up to 25,000 ft (7,620 m) MSL is identified as Category B2.

Category B3

Equipment intended for installation in the power plant compartment of an aircraft that is operated at altitudes up to 25,000 ft (7,620 m) MSL is identified as Category B3.

Category B4

Equipment intended for installation in a non-pressurized location on an aircraft that is operated at altitudes up to 25,000 ft (7,620 m) MSL, for which temperature requirements differ from B1and B2, is identified as Category B4.

Category C1

Equipment intended for installation in a non-pressurized but controlled temperature location in an aircraft that is operated at altitudes up to 35,000 ft (10,700 m) MSL is identified as Category C1.

Category C2

Equipment intended for installation in non-pressurized and non-controlled temperature locations within an aircraft that is operated at altitudes up to 35,000 ft (10,700 m) MSL is identified as Category C2.

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Category C3

Equipment intended for installation in the power plant compartment of an aircraft that is operated at altitudes up to 35,000 ft (10,700 m) MSL is identified as Category C3.

Category C4

Equipment intended for installation on a non-pressurized aircraft that is operated at altitudes up to 35,000 ft (10,700 m) MSL, for which temperature requirements differ from C1and C2, is identified as Category C4.

Category D1

Equipment intended for installation in a non-pressurized but controlled temperature location on an aircraft that is operated at altitudes up to 50,000 ft (15,200 m) MSL is identified as Category D1.

Category D2

Equipment intended for installation in non-pressurized and non-controlled temperature locations on an aircraft that is operated at altitudes up to 50,000 ft (15,200 m) MSL is identified as Category D2.

Category D3

Equipment intended for installation in the power plant compartment of an aircraft that is operated at altitudes up to 50,000 ft (15,200 m) MSL is identified as Category D3.

Category E1

Equipment intended for installation in non-pressurized and non-controlled temperature locations on an aircraft that is operated at altitudes up to 70,000 ft (21,300 m) MSL is identified as Category E1.

Category E2

Equipment intended for installation in the power plant compartment of an aircraft that is operated at altitudes up to 70,000 ft (21,300 m) MSL is identified as Category E2.

Category F1

Equipment intended for installation in non-pressurized but controlled temperature locations on an aircraft that is operated at altitudes up to 55,000 ft (16,800 m) MSL is identified as Category F1.

Category F2

Equipment intended for installation in non-pressurized and non-controlled temperature locations on an aircraft that is operated at altitudes up to 55,000 ft (16,800 m) MSL is identified as Category F2.

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Category F3

Equipment intended for installation in the power plant compartment of an aircraft that is operated at altitudes up to 55,000 ft (16,800 m) MSL is identified as Category F3.

4.4 Definitions of Terms

Operating Low Temperature

Operating low temperature is the lowest temperature at which equipment will normally be exposed and be required to operate.

Operating High Temperature

The operating high temperature values given in Table 4-1 are the maximum levels that the equipment will be exposed to within the particular installation area, e.g. in an enclosed space behind an instrument panel, equipment racks, power plant areas, etc., under normal operating conditions.

Short-Time Operating Temperature

These are startup conditions where equipment is turned on following a ground soak. It is expected that these temperature conditions will occur infrequently and be of short duration, since cooling or heating air circulation or other means of controlling temperature would be enabled concurrent with equipment operation.

Ground Survival Temperatures

These are the lowest and highest ground temperatures that the equipment is normally expected to be exposed to during aircraft storage or exposure to climatic extremes. The equipment is not expected to operate within specification limits at these temperatures but is expected to survive without damage.

In-Flight Loss of Cooling

This condition represents the failure of the external or internal system that normally provides dedicated cooling for the equipment. Certain equipment must survive for a limited time in the absence of cooling. Test requirements for this type of equipment shall be specified in the equipment performance specification.

4.5 Temperature Tests

4.5.1 Ground Survival Low Temperature Test and Short-Time Operating Low Temperature Test

At the ambient pressure and with the equipment not operating, stabilize the equipment at the appropriate ground survival low temperature specified in Table 4-1. Maintain this temperature for at least three hours. Then with the equipment not operating, increase the temperature to the

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short time operating low temperature specified in Table 4-1 at a minimum rate of 2°C/minute. Once the temperature has reached the short-time operating low temperature, soak the equipment for 30 +5/-0 minutes or until internal stabilization is established. Place the equipment into the operating state and maintain the test chamber air temperature at the appropriate short time operating low temperature specified in Table 4-1. Operate the equipment for at least 30 minutes. Verify equipment operation per note 1 during this operating period. The test profile is shown graphically in Figure 4-1.

Note: *1) This test simulates temperature conditions that may be encountered by equipment while the aircraft is on the ground. In determining the level of performance required during the period of this test, the operational requirements of the particular equipment or systems shall be stated in the test procedure and report or in the specific equipment performance standard.*

2) *If the short time operating low temperature and operating low temperature are the same, the short-time operating low temperature need not be conducted. The ground survival low temperature test may not be deleted, even if the short-time operating low temperature is identical to the operating low temperature. Power should be applied at the operating low temperature after the un-powered low temperature soak specified in paragraph 4.5.1.*

4.5.2 Operating Low Temperature Test

With the equipment operating, adjust the test chamber air temperature to the appropriate operating low temperature specified in Table 4-1 at ambient pressure. After the equipment temperature has become stabilized, operate the equipment for a minimum of two hours while maintaining the temperature of the air in the test chamber at the operating low temperature. DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS during this operating period. The test profile is shown graphically in figure 4-2.

4.5.3 Ground Survival High Temperature Test and Short-Time Operating High Temperature Test

At ambient pressure and with the equipment not operating, stabilize the equipment at the appropriate ground survival high temperature of Table 4-1. Maintain this temperature for at least three hours. Then with the equipment not operating, reduce the temperature to the short-time operating high temperature specified in Table 4-1 at a minimum rate of 2°C/minute. Once the temperature has reached the short-time operating high temperature, soak the equipment for 30 +5/-0 minutes or until internal stabilization is established. Place the equipment into the operating state and maintain the test chamber air temperature at the appropriate short-time operating high temperature specified in Table 4-1. Operate the equipment for at least 30 minutes. Verify equipment operation per note 1 during this operating period. The test profile is shown graphically in Figure 4-3.

Note: *1) This test simulates temperature conditions that may be encountered by equipment while the aircraft is on the ground. In determining the level of performance required during the period of this test, the operational requirements of the particular equipment or systems shall be stated in the test procedure and report or in the specific equipment performance standard.*

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2) If the short-time operating high temperature and operating high temperature are the same, the short-time operating high temperature test need not be conducted. The ground survival high temperature test may not be deleted, even if the short-time high temperature is identical to the operating high temperature. Power should be applied at the operating high temperature after the un-powered high temperature soak specified in paragraph 4.5.3.

4.5.4 Operating High Temperature Test

With the equipment operating, adjust the test chamber air temperature to the appropriate operating high temperature specified in Table 4-1 at ambient pressure. After the equipment temperature has become stabilized, operate the equipment for a minimum of two hours while maintaining the temperature of the air in the test chamber at the operating high temperature. DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS during the operating period. The test profile is shown graphically in Figure 4-4.

4.5.5 In-Flight Loss of Cooling Test

Categories of In-Flight Loss of Cooling Test Periods are defined by periods during which cooling is removed.

Category V - 30 minutes minimum
Category W - 90 minutes minimum
Category P - 180 minutes minimum
Category Y - 300 minutes minimum
Category Z - As defined in the equipment specification

With the equipment operating at ambient room pressure, and with cooling air supplied in accordance with the conditions specified in Subsection 3.4, adjust the test chamber air temperature to the value specified in Table 4-1 for the loss of cooling test, and allow the equipment temperature to stabilize. Turn off the equipment cooling air supply, and operate the equipment for the period of time specified for the applicable category while maintaining the temperature of the air in the test chamber at the value specified in Table 4-1. During this period DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS. The test profile is shown graphically in Figure 4-5.

Note: This test applies to equipment that requires cooling for proper operation during the operating high temperature test, paragraph 4.5.4, has functions whose failure following in-flight loss of cooling would contribute to or cause a failure condition that would prevent the continued safe flight and landing of the airplane, and in which the cooling is externally supplied or powered.

4.6 Altitude, Decompression and Overpressure Tests

Refer to Table 4-1 for altitude and to Table 4-2 for pressure values.

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4.6.1 Altitude Test

Conduct this test at ambient temperature. Operate the equipment at maximum duty cycle. Decrease the pressure in the test chamber to the appropriate maximum operating altitude specified in Table 4-1. Allow the equipment temperature to stabilize. Maintain this pressure for at least two hours. DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS during the two-hour period or at the maximum duty cycle, whichever is longer. The test profile is shown graphically in Figure 4-6.

Note: When the equipment manufacturer requires that the equipment be tested for spark-producing conditions at altitude, Sections 9.6a and 9.6b may apply. If so, the procedures of 9.7 shall be conducted at the maximum test altitude, and paragraph 3.3 (Combining Tests) may be applicable.

4.6.2 Decompression Test

Conduct this test at ambient temperature. With the equipment operating, adjust the absolute pressure to an equivalent altitude of 8,000 ft (2,400 m) MSL and allow the equipment temperature to stabilize. Reduce the absolute pressure to the equivalent of the maximum operational altitude for the aircraft on which the equipment will be installed (see Table 4-1). This reduction in pressure shall take place within 15 seconds. Maintain this reduced pressure for at least 10 minutes or as specified in the equipment specification. DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS during the period at maximum operating altitude. The test profile is shown graphically in Figure 4-7.

Note: The decompression test is intended for equipment as follows:

1) Equipment installed in pressurized areas on the aircraft required to operate during and following an emergency descent.

2) Equipment utilizing high voltage electrical/electronics circuits, i.e. displays etc...

Equipment intended for installation in areas that are subject to partial pressurization shall be tested in accordance with paragraph 4.6.1 above.

4.6.3 Overpressure Test

With the equipment not operating, unless otherwise specified in the equipment specification, subject the equipment to an absolute pressure equivalent to -15,000 ft altitude (170 kPa). Maintain this condition for at least 10 minutes. Return the equipment to the ambient atmospheric pressure and DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS. The test profile is shown graphically in Figure 4-8.

Note: 1) This test is for equipment installed in pressurized areas. The test determines whether the equipment will withstand cabin overpressures resulting from routine aircraft pressurization system testing.

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2) Equipment installed in a pressurized area and whose internal sections are vented external to the pressurized area shall have these internal sections exposed to the pressure specified in the equipment specification during the overpressure test.

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Table 4-1 Temperature and Altitude Criteria

Environmental Tests	Category Paragraph 4.3																			
	A				B				C				D			E		F		
	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	1	2	1	2	3
Operating Low Temp. Degrees C Paragraph 4.5.2	-15	-15	-15	-15	-20	-45	-45	Note (4)	-20	-55	-55	Note (4)	-20	-55	-55	-55	-55	-20	-55	-55
Operating High Temp. Degrees C Paragraph 4.5.4	+55	+70	+70	Note (3)	+55	+70	Note (3)	Note (4)	+55	+70	Note (3)	Note (4)	+55	+70	Note (3)	Note (3)	Note (3)	+55	+70	Note (3)
Short-Time Operating Low Temp. Degrees C Paragraph 4.5.1	-40	-40	-40	Note (3)	-40	-45	Note (3)	Note (4)	-40	-55	Note (3)	Note (4)	-40	-55	-55	-55	-55	-40	-55	-55
Short-Time Operating High Temp. Degrees C Paragraph 4.53	+70	+70	+85	Note (3)	+70	+70	Note (3)	Note (4)	+70	+70	Note (3)	Note (4)	+70	+70	Note (3)	Note (3)	Note (3)	+70	+70	Note (3)
Loss of Cooling Test Degrees C Paragraph 4.5.5	+30	+40	+45	Note (3)	+30	+40	Note (3)	Note (3)	+30	+40	Note (3)	Note (3)	+30	+40	Note (3)	Note (3)	Note (3)	+30	+40	Note (3)
Ground Survival Low Temperature Degrees C Paragraph 4.5.1	-55	-55	-55	Note (3)	-55	-55	Note (3)	-55	-55	-55	Note (3)	-55	-55	-55	-55	-55	-55	-55	-55	-55
Ground Survival High Temperature Degrees C Paragraph 4.5.3	+85	+85	+85	Note (3)	+85	+85	Note (3)	+85	+85	+85	Note (3)	+85	+85	+85	Note (3)	+85	Note (3)	+85	+85	Note (3)
Altitude Thousands of Feet Thousands of Meters Paragraph 4.6.1	15 4.6	15 4.6	15 4.6	15 4.6	25 7.6	25 7.6	25 7.6	25 7.6	35 10.7	35 10.7	35 10.7	35 10.7	50 15.2	50 15.2	50 15.2	70 21.3	70 21.3	55 16.8	55 16.8	55 16.8
Decompression Test Paragraph 4.6.2	Note (1) (4)	Note (1) (4)	Note (1) (4)	Note (1) (4)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Overpressure Test Paragraph 4.6.3	Note (2)	Note (2)	Note (2)	Note (2)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Note:

1) The lowest pressure applicable for the decompression test is the maximum operating altitude for the aircraft in which the equipment will be installed.
2) The absolute pressure is 170 kPa (-15,000 ft or -4,600 m).
3) To be declared by the equipment manufacturer relative to temperature extremes.
4) To be declared by the equipment manufacturer and defined in the manufacturer's installation instructions when specific critical criteria exist.

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Table 4-2 Pressure Values for Various Pressure Altitude Levels

Pressure Altitude	Absolute Pressure			
	kPa	(mbars)	(in Hg)	mm Hg
-15,000 ft (-4,572 m)	169.73	1697.3	50.12	1273.0
-1,500 ft (-457 m)	106.94	1069.4	31.58	802.1
0 ft (0m)	101.32	1013.2	29.92	760.0
+8,000 ft (+2,438 m)	75.26	752.6	22.22	564.4
+15,000 ft (+4,572 m)	57.18	571.8	16.89	429.0
+25,000 ft (+7,620 m)	37.60	376.0	11.10	282.0
+35,000 ft (+10,668 m)	23.84	238.4	7.04	178.8
+50,000 ft (+15,240 m)	11.60	116.0	3.42	87.0
+55,000 ft (+16,764 m)	9.12	91.2	2.69	68.3
+70,000 ft (+21,336 m)	4.44	44.4	1.31	33.3

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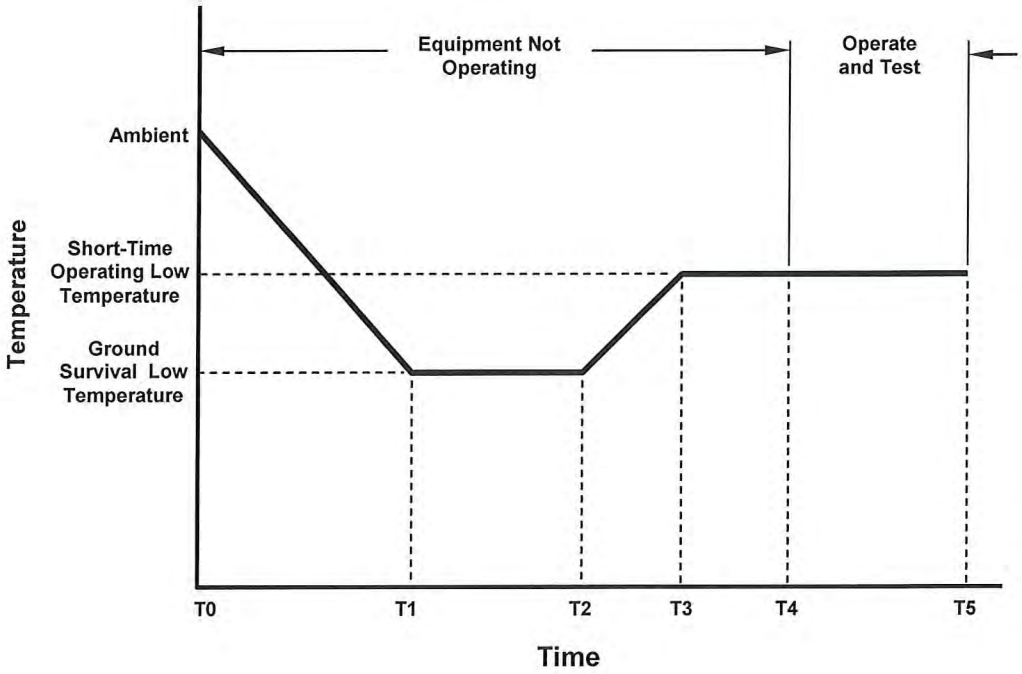

Note: *1) Temperature change rate from T0 to T1 is not specified.*

2) T1 to T2 is time for equipment temperature stabilization time, plus a minimum of three hours.

3) T2 to T3 is at a minimum rate of 2°C/minute

4) T3 to T4 is 30 +5/-0 minutes or the time for internal stabilization.

5) T4 to T5 is 0.5 hours, minimum.

6) If the Short-Time Low and Ground Survival Low Temperatures are identical, the time from T2 to T4 is zero.

7) See Note 2 of the test procedure if the short-time low operating temperature is the same as the operating low temperature.

Figure 4-1 **Ground Survival Low Temperature and Short-Time Operating Low Temperature Test**

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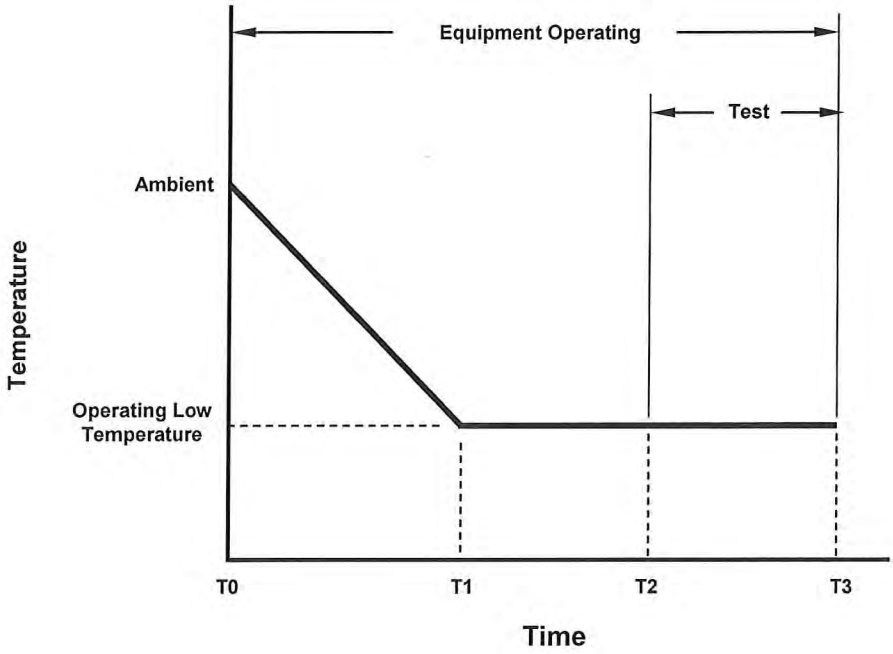

Note: 1) *Temperature change rate from T0 to T1 is not specified.*

2) *T1 to T2 is time for equipment temperature to stabilize.*

3) *T2 to T3 is 2.0 hours, minimum.*

Figure 4-2 Operating Low Temperature Test

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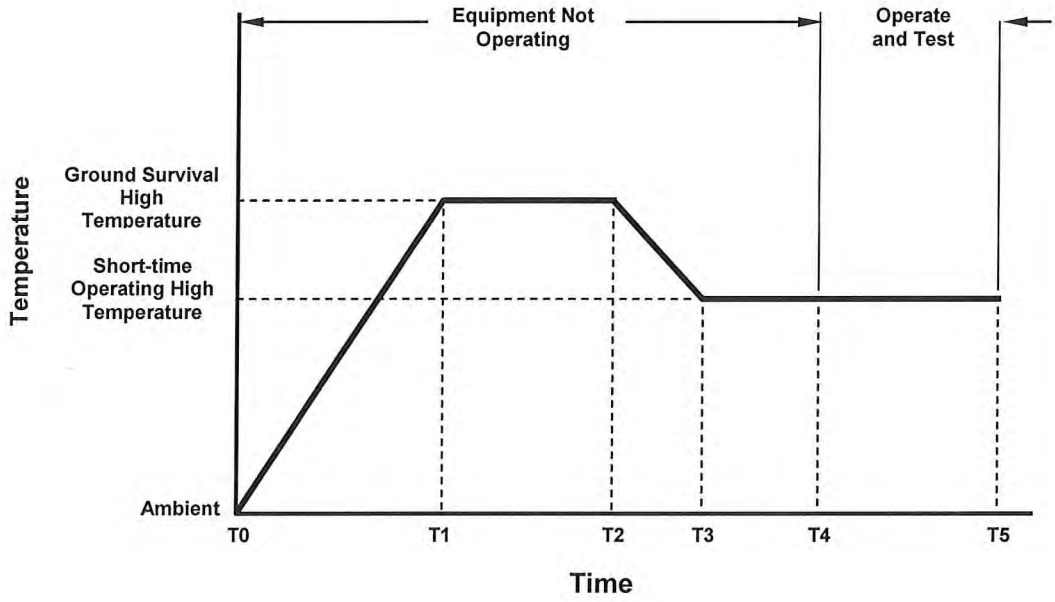

Note: *1) Temperature change rate from T0 to T1 is not specified.*

2) T1 to T2 is time for equipment temperature stabilization time, plus a minimum of three hours.

3) T2 to T3 is at a minimum rate of 2°C/minute

4) T3 to T4 is 30 +5/-0 minutes or the time for internal stabilization.

5) T4 to T5 is 0.5 hours, minimum.

6) If the short-time high and ground survival high temperatures are identical, the time from T2 to T4 is zero.

7) See Note 2 of the test procedure if the short-time high operating temperature is the same as the operating high temperature.

Figure 4-3 Ground Survival High Temperature and Short-Time Operating High Temperature Test

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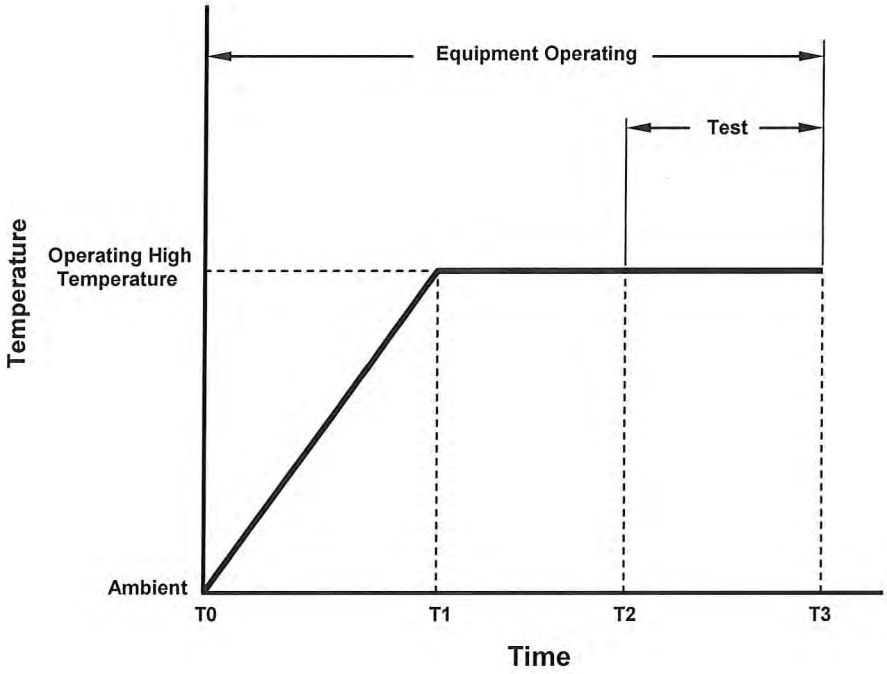

Note: 1) *Temperature change rate from T0 to T1 is not specified.*

2) *T1 to T2 is time for equipment temperature to stabilize.*

3) *T2 to T3 is 2.0 hours, minimum.*

Figure 4-4 Operating High Temperature Test

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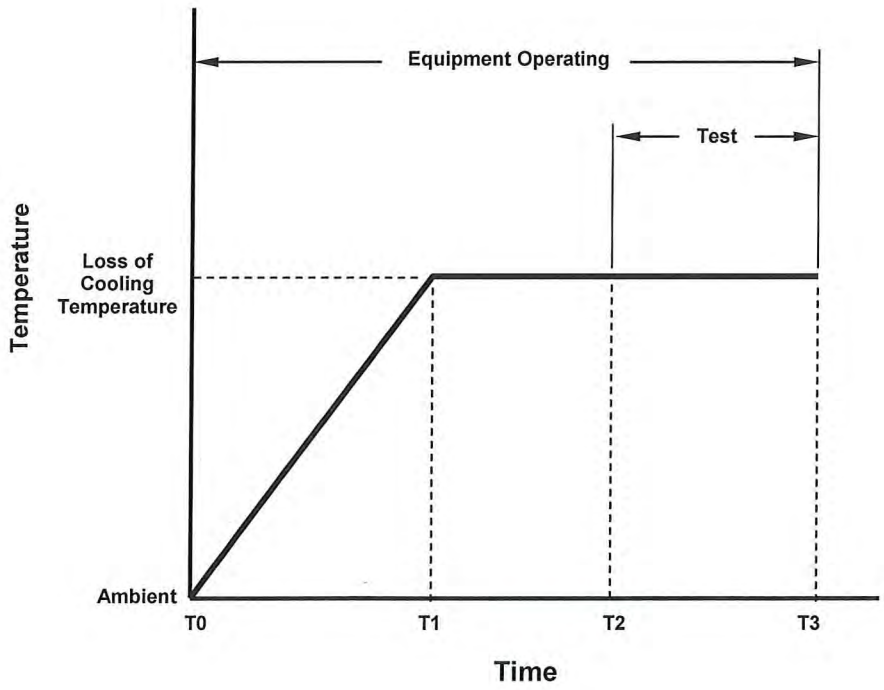

Note: 1) *Temperature change rate from T0 to T1 is not specified.*

2) *T1 to T2 is time for equipment temperature to stabilize.*

3) *See paragraph 4.5.4 for time duration T2 to T3.*

Figure 4-5 In-Flight Loss of Cooling Test

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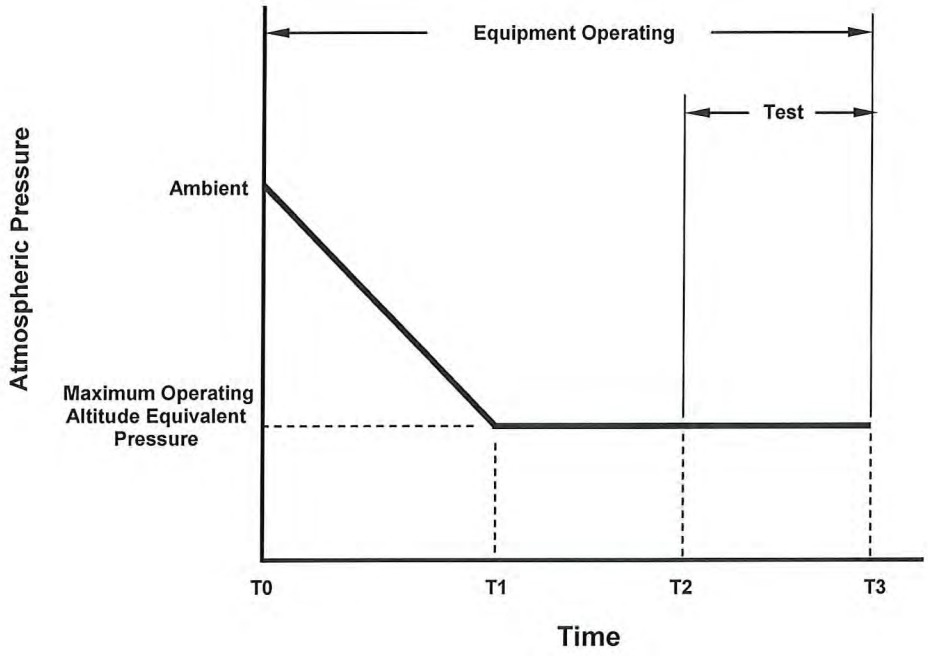

Note: *1) Pressure change rate from T0 to T1 is not specified.*

2) T1 to T2 is time for equipment temperature to stabilize.

3) T2 to T3 is 2.0 hours, minimum.

Figure 4-6 Altitude Test

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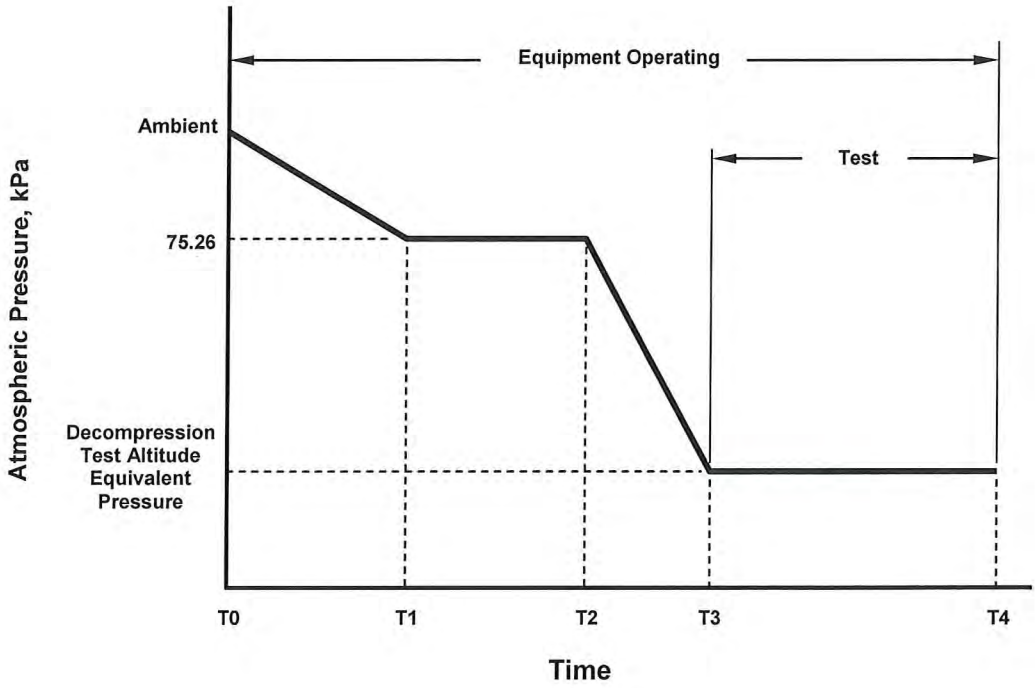

Note:

1) *Pressure change rate from T0 to T1 is not specified*
2) *T1 to T2 is time for equipment temperature to stabilize*
3) *T2 to T3 is 15 seconds, maximum*
4) *T3 to T4 is 10 minutes, minimum*

Figure 4-7 Decompression Test

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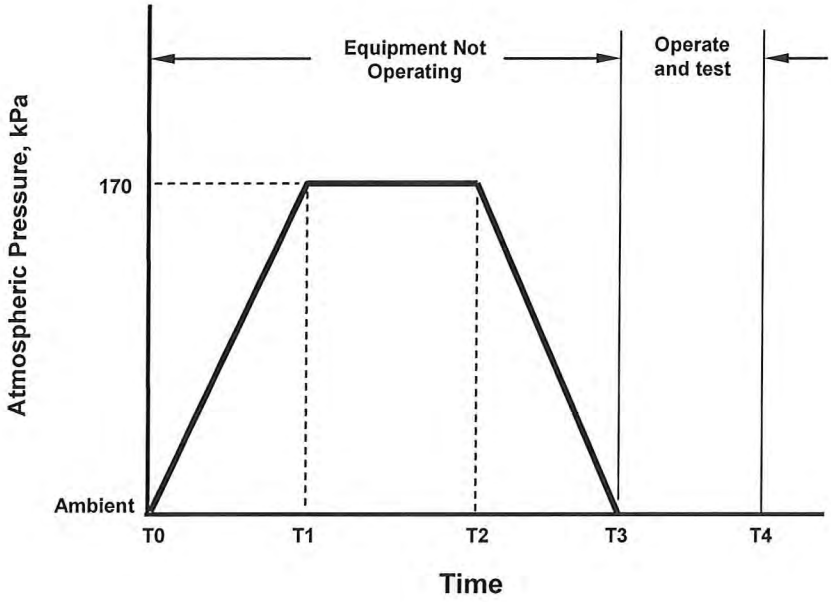

Note: 1) *Pressure change rates from T0 to T1 and from T2 to T3 are not specified.*

2) *T1 to T2 is 10 minutes, minimum.*

3) *T3 to T4 is minimum time necessary to operate and test equipment.*

Figure 4-8 Overpressure Test

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